



Exploring GAN Variants for Balancing Imbalanced Datasets

Using Breast Cancer Wisconsin Dataset

Course: Special Topics in Data Science

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1. Problem Statement

Imbalanced datasets represent a major challenge in machine learning applications, especially in medical diagnosis tasks. When one class contains significantly fewer samples than another, classification models tend to become biased toward the majority class, leading to poor detection of minority cases.

In the Breast Cancer Wisconsin dataset, malignant tumor samples are fewer than benign samples, creating a class imbalance problem. This imbalance can reduce the model's ability to correctly identify malignant tumors, which may lead to dangerous false negative predictions in medical applications.

The purpose of this project is to address the imbalance problem using Generative Adversarial Networks (GANs). Vanilla GAN and Wasserstein GAN (WGAN) were implemented to generate synthetic malignant samples and balance the dataset. The performance of classification models was then compared before and after balancing using multiple evaluation metrics such as Accuracy, Precision, Recall, F1-score, and Confusion Matrix.

2. Dataset & Imbalance Analysis

2.1 Dataset Overview

The Breast Cancer Wisconsin dataset was used in this project. The dataset contains medical measurements extracted from digitized images of breast cell nuclei obtained through Fine Needle Aspiration (FNA).

The objective of the dataset is to classify tumors into two categories:

- Benign (B)
- Malignant (M)

The dataset contains:

- 569 total samples
- 30 numerical features describing characteristics of the cell nuclei
- The target variable is the diagnosis column, where:
 - Malignant = 1
 - Benign = 0

The dataset does not contain missing values, which simplifies the preprocessing stage and improves model training efficiency.

Dataset Link: [Breast Cancer Wisconsin \(Diagnostic\) Data Set](#)

2.2 Class Imbalance

The dataset contains an imbalanced distribution between benign and malignant tumor samples:

- Benign samples: 357
- Malignant samples: 212

Since malignant samples represent the minority class, this imbalance may negatively affect classification performance, especially for detecting malignant tumors.

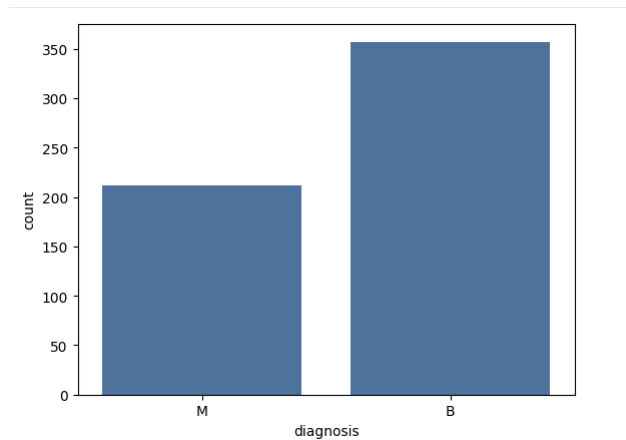


Figure 1: Class Distribution of the Breast Cancer Wisconsin Dataset

3. GAN Architectures & Training

3.1 Vanilla GAN

A Vanilla Generative Adversarial Network (GAN) was implemented to generate synthetic samples for the minority class (malignant tumors).

The Vanilla GAN architecture consists of two neural networks:

- Generator: generates synthetic samples from random noise.
- Discriminator: distinguishes between real and generated samples.

Both networks are trained simultaneously using adversarial learning, where the generator tries to fool the discriminator while the discriminator attempts to correctly classify real and fake samples.

The model was trained on the minority class samples for 1000 epochs to generate synthetic malignant data and reduce the class imbalance problem.

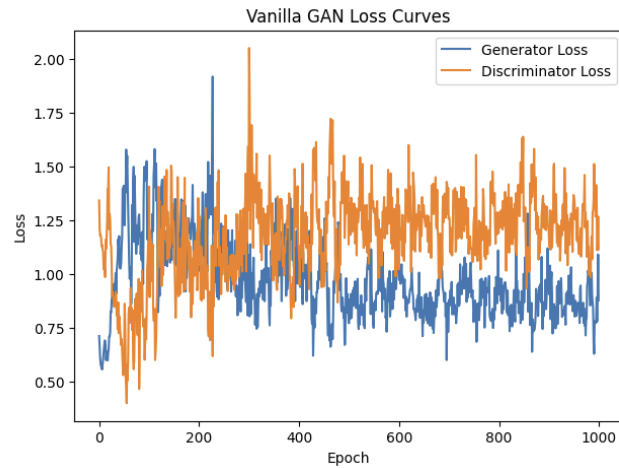


Figure 2: Vanilla GAN training loss curves showing Generator and Discriminator losses during training.

3.2 Wasserstein GAN (WGAN)

A Wasserstein Generative Adversarial Network (WGAN) was implemented as an improved GAN variant to generate synthetic malignant tumor samples.

Unlike Vanilla GAN, WGAN replaces the discriminator with a critic network and uses Wasserstein distance as the loss function. This approach improves training stability and reduces common GAN training problems such as unstable convergence and mode collapse.

The WGAN model was trained on the minority class samples for 1000 epochs. The generated synthetic samples were then used to balance the dataset and improve classification performance.

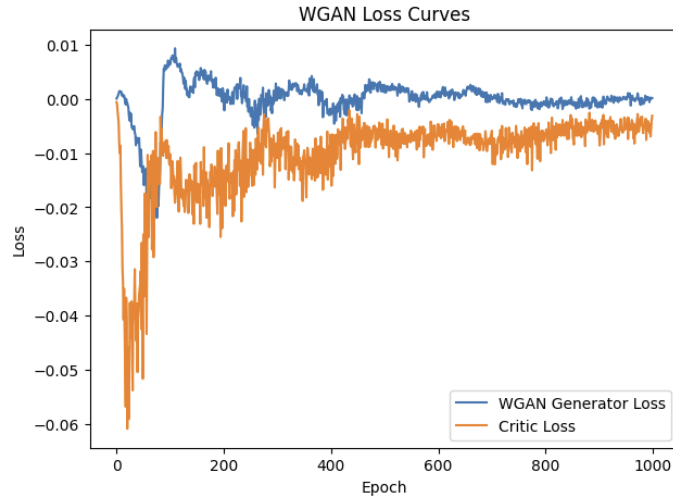


Figure 3: WGAN training loss curves showing Generator and Critic losses during training.

4. Classifier Setup and Evaluation

4.1 Classifier

An MLP (Multi-Layer Perceptron) classifier was used to evaluate the classification performance of the dataset before and after balancing.

The classifier was trained on three different datasets:

1. Original imbalanced dataset
2. Balanced dataset generated using Vanilla GAN
3. Balanced dataset generated using WGAN

The dataset was divided into training and testing sets using an 80/20 split. The classifier performance was evaluated using several metrics, including:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix

The objective of this stage was to analyze how balancing the dataset using GAN-generated samples affects classification performance, especially for detecting malignant tumors.

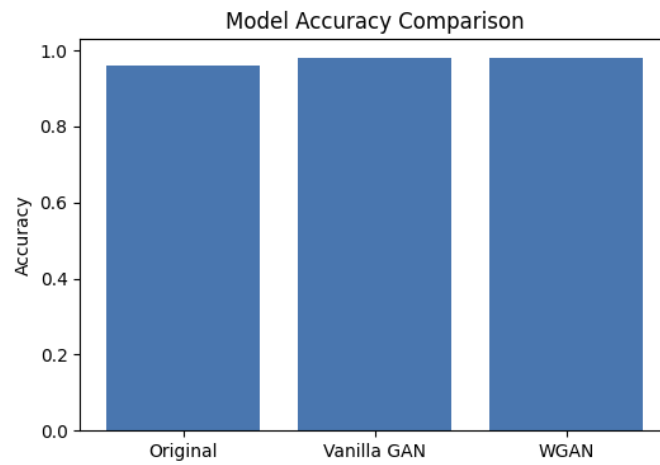


Figure 4: Accuracy comparison between the original dataset, Vanilla GAN balanced dataset, and WGAN balanced dataset.

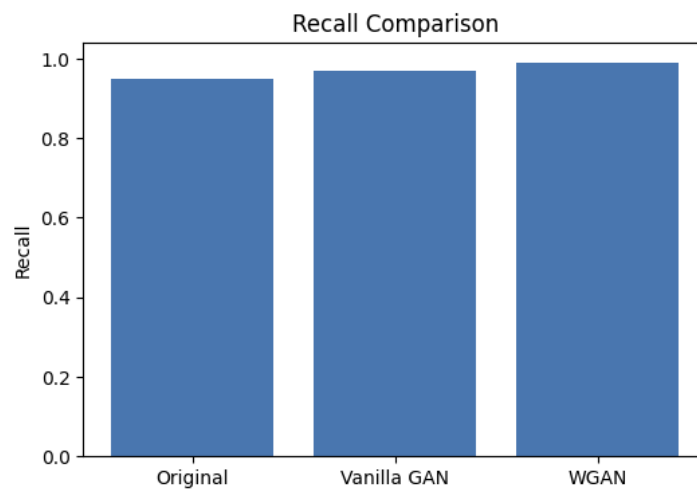


Figure 5: Recall comparison showing improved detection of malignant tumors after dataset balancing.

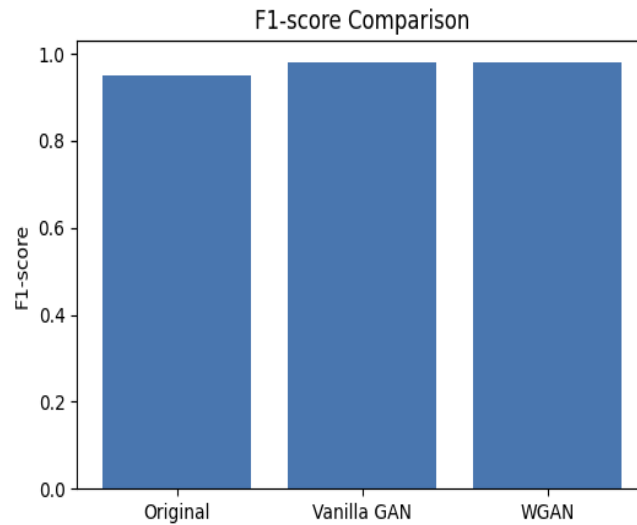


Figure 6: F1-score comparison between classification models trained on different datasets.

4.2 Experimental Scenarios

Three experimental scenarios were used to evaluate the effect of GAN-based balancing techniques on classification performance:

1. Original Imbalanced Dataset

The classifier was trained using the original dataset without any balancing techniques.

2. Balanced Dataset Using Vanilla GAN

Synthetic malignant samples generated by Vanilla GAN were added to the dataset to create a balanced class distribution before classification.

3. Balanced Dataset Using WGAN

Synthetic malignant samples generated by WGAN were used to balance the dataset and evaluate whether WGAN improves classification performance compared to Vanilla GAN.

The performance of all scenarios was compared using Accuracy, Precision, Recall, F1-score, and Confusion Matrix.

5. Results & Comparisons

5.1 Classification Metrics

The classification performance was evaluated using Accuracy, Precision, Recall, and F1-score for the three experimental scenarios.

The results are summarized in Table 1.

Metric	Original	Vanilla GAN	WGAN
Accuracy	96%	98%	98%
Precision	95%	99%	97%
Recall	95%	97%	99%
F1-Score	95%	98%	98%
AUC-ROC	0.9948	0.9971	0.9975

The results show that balancing the dataset using GAN-generated samples improved the overall classification performance compared to the original imbalanced dataset.

Vanilla GAN achieved the highest Precision, while WGAN achieved the highest Recall, indicating better detection of malignant tumors. Since Recall is especially important in medical diagnosis tasks, WGAN provided the most effective performance for detecting malignant cases.

5.2 Key Observations

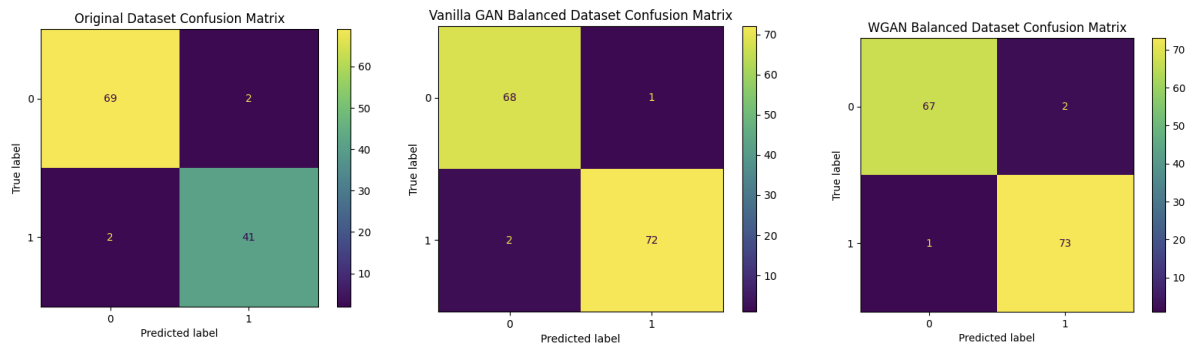
- GAN-based balancing improved classification performance compared to the original imbalanced dataset.
- Vanilla GAN achieved the highest Precision, while WGAN achieved the highest Recall for detecting malignant tumors.

- WGAN training was more stable than Vanilla GAN, as shown in the loss curves.
- Overall, WGAN provided the best performance for handling the imbalanced breast cancer dataset.

5.3 Confusion Matrix Summary

The confusion matrices showed that the balanced datasets generated using GANs reduced classification errors compared to the original imbalanced dataset.

WGAN achieved fewer false negatives and better detection of malignant tumors, which is especially important in medical diagnosis applications. The results confirm that balancing the dataset using synthetic samples improved the classifier's performance.



6. Observations and Conclusions

6.1 Observations

The experimental results showed that balancing the dataset using GAN-generated samples improved classification performance compared to the original imbalanced dataset.

Both Vanilla GAN and WGAN increased Accuracy, Recall, and F1-score. WGAN achieved the best Recall and showed more stable training behavior, making it more effective for detecting malignant tumors in the breast cancer dataset.

6.2 Conclusions

This project demonstrated the effectiveness of GAN-based techniques for handling imbalanced datasets in medical classification tasks.

Both Vanilla GAN and WGAN improved classification performance by generating synthetic malignant samples and balancing the dataset. Among the tested approaches, WGAN achieved the best overall performance due to its stable training behavior and higher Recall for detecting malignant tumors.

The results confirm that GAN-based data augmentation can significantly improve classification models when dealing with imbalanced medical datasets.